JAVA PROGRAMS 06-03-26

1

```
package project;

import java.util.Arrays;
import java.util.Scanner;

public class SortingNumbers {
	static Scanner sc = new Scanner(System.in);
	public static void main(String[] args) {
		// TODO Auto-generated method stub
		// int arr[] =
{15,9,4,6,3,1,2,11,10,13,14,12,5,7,8};
		int arr[] = new int[15];
		System.out.print("Enter 15 number:\t");
		for (int i = 0; i < arr.length; i++) {
			arr[i] = sc.nextInt();
		}
		System.out.println("Original
Array:\t"+Arrays.toString(arr));
		for (int i = 0; i < arr.length; i++) {
			Arrays.sort(arr);
		}

		System.out.println("Numbers sorted in Ascending
Order: ");
		System.out.println(Arrays.toString(arr));
		// System.out.println(arr);
	}
}
```

Output:

```
Enter 15 number:	13 14 12 11 15 7 5 6 4 2 1 3 9 8 10
Original Array:	[13, 14, 12, 11, 15, 7, 5, 6, 4, 2, 1,
3, 9, 8, 10]
Numbers sorted in Ascending Order:
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]
```

2.

```java
package project;

import java.util.Scanner;

public class ReverseString {
     static Scanner sc = new Scanner(System.in);

     public static void main(String[] args) {
          System.out.print("Enter a sentence: \t");
          String sentence = sc.nextLine();
          String rev = "";
          for (int i = sentence.length() - 1; i >= 0; i--)
{
               rev = rev + sentence.charAt(i);
          }
          System.out.println(rev);
     }
}
```

Output:

```
Enter a sentence:  Java Programming
gnimmargorP avaJ
```

3.

```
package project;

import java.util.*;

public class DenominationValues {
     static Scanner sc = new Scanner(System.in);

     public static void main(String[] args) {
          // TODO Auto-generated method stub
          System.out.print("Enter a Value with long digits:
\t");
          long value_number = sc.nextLong();
          System.out.println("How many Rs 500 can be there
in it: " + value_number / 500);
          System.out.println("How many Rs 100 can be there
in it: " + value_number / 100);
          System.out.println("How many Rs 50 can be there
in it: " + value_number / 50);
          System.out.println("How many Rs 20 can be there
in it: " + value_number / 20);
          System.out.println("How many Rs 10 can be there
in it: " + value_number / 10);
          System.out.println("How many Rs 1 can be there in
it: " + value_number / 1);
     }

}
```

Output:

```
Enter a Value with long digits:  1000
How many Rs 500 can be there in it: 2
How many Rs 100 can be there in it: 10
How many Rs 50 can be there in it: 20
How many Rs 20 can be there in it: 50
How many Rs 10 can be there in it: 100
How many Rs 1 can be there in it: 1000
```

4.

```java
package project;

public class CharacterMatch {
    static String sentence = "welcome to programming";
    static char chars[] = sentence.toCharArray();
    static int n = sentence.length();
    static int count0 = 0;
    static int count1 = 0;
    static int count2 = 0;
    static int count3 = 0;
    static int count4 = 0;
    static int count5 = 0;
    static int count6 = 0;
    static int count7 = 0;
    static int count8 = 0;
    static int count9 = 0;
    static int count10 = 0;
    static int count11 = 0;
    static int count12 = 0;
    static int count13 = 0;
    static int count14 = 0;
    static int count15 = 0;
    static int count16 = 0;
    static int count17 = 0;
    static int count18 = 0;
    static int count19 = 0;
    static int count20 = 0;
    static int count21 = 0;
    static int count22 = 0;
    static int count23 = 0;
    static int count24 = 0;
    static int count25 = 0;
    static int count26 = 0;
    // static int i;

    public static void main(String[] args) {
        for (int i = 0; i < n; i++) {
            if (chars[i] == 'a') {
                count0++;
```

```
                System.out.println("Count of A is: " +
count0);
            } else if (chars[i] == 'b') {
                count1++;
                System.out.println("Count of B is: " +
count1);
            } else if (chars[i] == 'c') {
                count2++;
                System.out.println("Count of C is: " +
count2);
            } else if (chars[i] == 'd') {
                count3++;
                System.out.println("Count of D is: " +
count3);
            } else if (chars[i] == 'e') {
                count4++;
                System.out.println("Count of E is: " +
count4);
            } else if (chars[i] == 'f') {
                count5++;
                System.out.println("Count of F is: " +
count5);
            } else if (chars[i] == 'g') {
                count6++;
                System.out.println("Count of G is: " +
count6);
            } else if (chars[i] == 'h') {
                count7++;
                System.out.println("Count of H is: " +
count7);
            } else if (chars[i] == 'i') {
                count8++;
                System.out.println("Count of I is: " +
count8);
            } else if (chars[i] == 'j') {
                count9++;
                System.out.println("Count of J is: " +
count9);
            } else if (chars[i] == 'k') {
                count10++;
                System.out.println("Count of K is: " +
count10);
```

```
                } else if (chars[i] == 'l') {
                    count11++;
                    System.out.println("Count of L is: " +
count11);
                } else if (chars[i] == 'm') {
                    count12++;
                    System.out.println("Count of M is: " +
count12);
                } else if (chars[i] == 'n') {
                    count13++;
                    System.out.println("Count of N is: " +
count13);
                } else if (chars[i] == 'o') {
                    count14++;
                    System.out.println("Count of O is: " +
count14);
                } else if (chars[i] == 'p') {
                    count15++;
                    System.out.println("Count of P is: " +
count15);
                } else if (chars[i] == 'q') {
                    count16++;
                    System.out.println("Count of Q is: " +
count16);
                } else if (chars[i] == 'r') {
                    count17++;
                    System.out.println("Count of R is: " +
count17);
                } else if (chars[i] == 's') {
                    count18++;
                    System.out.println("Count of S is: " +
count18);
                } else if (chars[i] == 't') {
                    count19++;
                    System.out.println("Count of T is: " +
count19);
                } else if (chars[i] == 'u') {
                    count20++;
                    System.out.println("Count of U is: " +
count20);
                } else if (chars[i] == 'v') {
                    count21++;
```

```
                    System.out.println("Count of V is: " +
count21);
                } else if (chars[i] == 'w') {
                    count22++;
                    System.out.println("Count of W is: " +
count22);
                } else if (chars[i] == 'x') {
                    count23++;
                    System.out.println("Count of X is: " +
count23);
                } else if (chars[i] == 'y') {
                    count24++;
                    System.out.println("Count of Y is: " +
count24);
                } else if (chars[i] == 'z') {
                    count25++;
                    System.out.println("Count of Z is: " +
count25);
                }
            }
        }
}
```

Output:

```
Count of W is: 1
Count of E is: 1
Count of L is: 1
Count of C is: 1
Count of O is: 1
Count of M is: 1
Count of E is: 2
Count of T is: 1
Count of O is: 2
Count of P is: 1
Count of R is: 1
Count of O is: 3
Count of G is: 1
Count of R is: 2
Count of A is: 1
Count of M is: 2
Count of M is: 3
```

```
Count of I is: 1
Count of N is: 1
Count of G is: 2
```

5.

```
package project;

import java.util.Scanner;
public class TeamMatchesF2F {
     static Scanner sc = new Scanner(System.in);
     public static void main(String[] args) {
          // TODO Auto-generated method stub
              System.out.print("Enter number of teams: ");
             int teams = sc.nextInt();

             System.out.print("Enter how many times each
team plays against another: ");
             int times = sc.nextInt();

             int totalMatches = (teams * (teams - 1) / 2)
* times;

             System.out.println("Total number of matches
to be played: " + totalMatches);

             sc.close();
     }
}
```

Output:

```
Enter number of teams: 2
Enter how many times each team plays against another: 1
Total number of matches to be played: 1
```

6.

```java
package project;

import java.util.*;

public class Palindrome {
	static Scanner sc = new Scanner(System.in);

	public static void main(String[] args) {
		// TODO Auto-generated method stub
		System.out.print("Enter a string:\t");
		String str = sc.nextLine().toLowerCase();
		int n = str.length();
		String reversed = "";
		for (int i = n - 1; i >= 0; i--) {
			reversed = reversed + str.charAt(i);
		}
		if(reversed.equals(str)) {
			System.out.println("Palindrome");
		}
		else
			System.out.println("Not a Palindrome");
		sc.close();
	}
}
```

Output:

```
Enter a string:	Madam
Palindrome
```

7.

```
package project;

import java.util.Arrays;
import java.util.Scanner;

public class NumberRepetitionCount {
    static Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements in
the array: ");
        int size = sc.nextInt();
        int[] arr = new int[size];
        System.out.println("Enter array elements:");
        for (int i = 0; i < size; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.println("Array Elements: " +
Arrays.toString(arr));
        for (int i = 0; i < size; i++) {
            int count = 0;
            boolean alreadyCounted = false;
            for (int k = 0; k < i; k++) {
                if (arr[i] == arr[k]) {
                    alreadyCounted = true;
                    break;
                }
            }
            if (alreadyCounted) {
                continue;
            }
            for (int j = 0; j < size; j++) {
                if (arr[i] == arr[j]) {
                    count++;
                }
            }
            System.out.println(arr[i] + " was entered "
+ count + " times.");
        }
```

```
        sc.close();
    }
}
```

Output:

```
Enter the number of elements in the array: 5
Enter array elements:
1 1 1 2 3
Array Elements: [1, 1, 1, 2, 3]
1 was entered 3 times.
2 was entered 1 times.
3 was entered 1 times.
```

8.

```
package project;

import java.util.*;

public class CurrentAge_Year {
     static Scanner sc = new Scanner(System.in);
     public static void main(String[] args) {
          // TODO Auto-generated method stub
          int year = 2026;
          System.out.println("Enter the age of the person:
");
          int age = sc.nextInt();
          int Yearofbirth = year - age;
          if (age > 0) {
               System.out.println("year of birth is: " +
Yearofbirth);
          }
          else {
               System.out.println("Check your input age
value");
          }
     }
}
```

Output:

```
Enter the age of the person:
22
year of birth is: 2004
```

9.

```java
package project;

import java.util.Scanner;

public class NumberTriangle {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        for (int i = 1; i <= rows; i++) {
            for (int j = 1; j <= rows - i; j++) {
                System.out.print("   ");
            }
            for (int num = 1; num <= i; num++) {
                System.out.print(num + "   ");
            }
            System.out.println();
        }
        sc.close();
    }
}
```

Output:

```
Enter number of rows: 5
            1
         1   2
      1   2   3
   1   2   3   4
1   2   3   4   5
```

10.

```java
package project;

import java.util.Arrays;
import java.util.Scanner;

public class BiggestNumber {
    static Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {

        int numbers[] = new int[5];
        int max = Integer.MIN_VALUE;
        int n = numbers.length;

        System.out.print("Enter 5 numbers:\t");

        for (int i = 0; i < n; i++) {
            numbers[i] = sc.nextInt();
        }

        System.out.println("The Array Elements
are:\t"+Arrays.toString(numbers));

        for (int i = 0; i < n; i++) {
            if (numbers[i] > max) {
                max = numbers[i];
            }
        }
        System.out.println("Maximum element among the
number:\t" + max);
    }
}
```

Output:

```
Enter 5 numbers:   1 2 3 4 5
The Array Elements are:[1, 2, 3, 4, 5]
Maximum element among the number:     5
```